Evaluating Heuristic Post-Hoc Explanations Against Formal Sufficiency Standards

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***Abstract*—This paper evaluates the accuracy of LIME, SHAP and Anchors by treating logic-based sufficient reasons as ground truth explanations on two UCI benchmark datasets (Adult Income and German Credit). Sufficient reasons are computed via SAT-based enumeration, and several methods for converting heuristic outputs to feature subsets are tested, including top-k by absolute value, cumulative contribution, and threshold-based selection. Using top-k absolute value ranking, SHAP reaches 99% sufficiency on Adult Income at a budget of seven features, while LIME climbs more slowly to 89% at nine features and Anchors remains near 0%. To measure the repair cost of the heuristic explanations, we extend each method's own ranking until a sufficiency check passes and count the features added. On Adult Income SHAP needs almost none, but on the higher-dimensional German Credit, where sufficient reasons average eight features, sufficiency stays low even at a nine-feature budget and the repair cost stays high. Sufficiency therefore degrades sharply with feature dimensionality.**

***Index Terms*—Explainable AI, sufficient reasons, LIME, SHAP, Anchors, decision trees**

I. Introduction

Post-hoc explanation methods for machine learning can be broadly divided into two categories. Heuristic methods such as LIME [4] and SHAP [5] produce feature importance scores or contribution values, measuring how much each feature influenced a prediction. Logic-based methods such as those formalised by Darwiche and Hirth [1] produce sufficient reasons: minimal sets of feature conditions that formally guarantee the prediction regardless of all other features. These are fundamentally different outputs: a SHAP value tells us how much a feature contributed while a sufficient reason indicates that fixing certain features locks in the prediction. In practice, however, practitioners routinely interpret the top-ranked LIME or SHAP features as "the reason" for a prediction. In high-stakes domains such as credit scoring and medical diagnosis, the gap between what heuristic methods measure and what formal frameworks guarantee has direct consequences for the trustworthiness of explanations.

Existing evaluations either classify explanations as simply correct or incorrect without analysing why they fail or what it would cost to make them sufficient [3], or are limited to small synthetic settings [7]. By cost, we mean the following: when a heuristic explanation is not sufficient, it can be *repaired* by adding more of the features the heuristic ranks highly until the resulting subset formally guarantees the prediction, and the number of features added measures how far short the original explanation fell. This thesis evaluates LIME, SHAP, and Anchors on real tabular datasets, introduces diagnostic metrics that decompose the gap between heuristic outputs and formal sufficiency, and measures this repair cost.

The main research question guiding this work is: *To what extent can heuristic post-hoc explanation methods satisfy the formal criteria of sufficient reasons from logic-based explanation frameworks?*

This is addressed through three sub-questions:

RQ1 How can the heuristic outputs of LIME, SHAP, and Anchors be translated into feature subsets for comparison against logic-based sufficient reasons?

RQ2 To what extent do the resulting feature subsets satisfy sufficiency?

RQ3 What is the practical cost of repairing heuristic explanations to satisfy formal criteria, and does this differ across methods?

Our contributions are an evaluation of sufficiency rates across budgets k=1–9 on two UCI datasets, diagnostic metrics (overlap, waste, Jaccard similarity and repair cost) for the gap between heuristic and logic-based explanations, a comparison of several translation methods, and a LIME hyperparameter sensitivity analysis.

Section II covers background, section III covers methodology, section IV covers results, section V covers discussion, and section VI concludes the paper.

II. Background

A. Formal Explanations

A *sufficient reason* for a decision formalises the idea that certain characteristics of an instance are enough to guarantee the decision, regardless of all other characteristics. The concept originates from Shih et al. [15], who introduced PI-explanations (prime implicant explanations) for Bayesian network classifiers. Ignatiev et al. [2] proposed the equivalent concept of abductive explanations, and Darwiche and Hirth [1] unified these under the term sufficient reason, which we adopt throughout this paper.

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AI tools were used for parts of the implementation.

We first define the terminology used below. A classifier is treated as a Boolean formula Δ whose models (assignments that satisfy the formula) are the inputs it assigns to a given class, so that a decision can be reasoned about symbolically. An *instance* α is a complete assignment of values to all features, written as a conjunction of literals (one literal per feature, such as age ≤ 45.5). A *property* of an instance is a subset of those literals, that is, a partial description that α satisfies. A term (a conjunction of literals) is an *implicant* of Δ if every instance satisfying the term is classified the same way by Δ; it is a *prime implicant* if no literal can be removed from it without breaking this guarantee. With these in place, the definition is as follows.

Definition 1 (Sufficient Reason [1], [15]). A sufficient reason for decision $\Delta(\alpha)$ is a property of instance α that is also a prime implicant of Δ_α, where $\Delta_\alpha = \Delta$ if the decision is positive and $\Delta_\alpha = \neg\Delta$ if the decision is negative.

In words, a sufficient reason is a minimal subset of the instance's feature values that forces a certain prediction: fixing those values guarantees the outcome no matter what the remaining features are, and no value in the subset is redundant. A decision can have several sufficient reasons, and Darwiche and Hirth [1] (Proposition 1) prove that every decision has at least one.

This definition assumes only that the classifier can be written as a formula Δ and characterises sufficient reasons as prime implicants, but it does not say how the classifier is encoded or how the reasons are computed in practice. For a concrete representation, both can be made specific. For a decision tree, a sufficient reason is a minimal subset of the feature conditions along the instance's prediction path such that, once those features are fixed to their instance values, every assignment of the remaining features reaches a leaf with the same label. To compute these, we follow Ignatiev et al. [2]: start from all features on the decision path and tentatively remove each one, re-checking sufficiency via a SAT solver. If removing a feature preserves sufficiency it is discarded, otherwise it is kept, and the result is subset-minimal. All reasons are enumerated using blocking clauses that prevent previously found reasons from being returned again. The PyXAI [13] Python library implements this procedure using a Glucose SAT solver.

B. Heuristic Explanation Methods

LIME [4] explains a prediction by fitting a local surrogate model. For a given instance, it generates random perturbations, obtains the classifier's predictions for each, and fits a weighted Lasso regression on the resulting local dataset. The coefficients of this linear model are the *feature importance scores*: one signed number per feature whose magnitude indicates how strongly that feature drives the local prediction and whose sign indicates the direction. LIME is an additive attribution method, meaning the fitted scores sum to the surrogate model's output for the instance. The `num_features` parameter specifies how many features the local model retains, and each value of k (the number of features to select) produces a separately fitted model, so the feature ranking can change between different values of k.

SHAP [5] assigns each feature a *contribution score* based on Shapley values from coalitional game theory. A feature's *marginal contribution* to a coalition is the change in the model's output when that feature is added to it, and the Shapley value averages this marginal contribution over all possible coalitions of the other features. Like LIME, SHAP is additive: the contribution scores sum to the difference between the model's output for the instance and its expected output. For tree-based models, the SHAP library's TreeExplainer module uses TreeSHAP, which computes exact Shapley values in polynomial time by traversing the tree structure [16]. The resulting feature ranking is fixed and deterministic for a given instance.

Anchors [6] produces IF-THEN rules via beam search, targeting a user-specified precision threshold τ (default 0.95). Precision is estimated by sampling perturbations and checking whether the prediction is stable. The rule A satisfies $E_{\mathcal{D}(z|A)}[\mathbf{1}_{f(x)=f(z)}] \geq \tau$, where $\mathcal{D}(z|A)$ is the distribution of instances matching the Anchors conditions. Unlike LIME and SHAP, Anchors directly outputs a feature subset and determines its own explanation size.

C. The Translation Problem

The standard practitioner interpretation of feature importance and feature contribution scores may be to select the top-k features by absolute score magnitude, which reflects how SHAP waterfall plots and LIME bar charts are typically consumed. We refer to this as the *top-k translation* with *absolute value ranking*.

To address RQ1 more broadly, we also evaluate alternative translations grounded in the literature: cumulative contribution selection based on the efficiency property of Shapley values [5], threshold-based selection inspired by shap-select [12], Formal Feature Attribution (FFA) rankings based on feature frequency across all sufficient reasons [11], and signed ranking as recommended by Salem [7]. These experiments are detailed in Section III-E.

D. Related Work

Ignatiev [3] evaluated LIME, SHAP, and Anchors on boosted trees across five benchmark datasets using a SAT-based validation framework. Explanations were classified as correct, redundant, or incorrect (Figure 2 in [3]). Table 1 of that paper reports that for 4 out of 5 datasets, correct explanations constitute less than 31% for LIME, 20% for SHAP, and 18% for Anchors. Our work uses a simpler model (single decision trees with exact tree traversal rather than boosted trees with SAT encoding) and adds diagnostic metrics that decompose *why* explanations fail and *what it costs* to repair them.

Salem [7] evaluated the same three methods on synthetic Boolean data with 4–8 features, defining alignment conditions based on feature relevancy, necessity, and frequency. Results showed SHAP aligns well on simple Boolean functions and that LIME's `num_samples` parameter affects alignment.

Future work recommended evaluation on more complex real-world data and analysis of the sign of contribution values. Our work addresses both recommendations: we use real tabular datasets with 14–20 features, evaluate the sufficiency rate of derived feature subsets, test signed vs. absolute ranking, and analyse LIME hyperparameter sensitivity on real data.

Mothilal et al. [8] used the actual causality framework to define probabilistic necessity and sufficiency scores for top-k feature sets, finding that LIME and SHAP features are "often neither necessary nor sufficient" on Adult Income and German Credit. Their sufficiency is distributional, measuring the probability that the prediction holds when non-subset features are sampled from a reference distribution. Ours is formal, requiring the prediction to hold for all possible values of free features, and is computed exactly via tree traversal rather than estimated via sampling.

Gorji and Rubin [9] showed that incorporating domain constraints into sufficiency checking yields more parsimonious reasons, and that ignoring constraints produces unnecessarily long explanations. This motivates our encoding choice (Section III-A).

III. Methodology

A. Datasets and Model

We use two UCI benchmark datasets: Adult Income (14 features, 30,162 instances) for predicting whether annual income exceeds \$50K, and German Credit (20 features, 1,000 instances) for predicting creditworthiness. Both are standard benchmarks in the XAI evaluation literature [3], [6], [8].

Categorical features are integer-encoded: each category is mapped to a sequential integer based on alphabetical sort order. This avoids the mutual-exclusivity constraint problem identified by Gorji and Rubin [9]: with one-hot encoding, a sufficiency checker can assign conflicting indicator values simultaneously, making sufficiency checks overly strict. We acknowledge as a limitation the trade-off that integer encoding imposes an ordinal assumption on categorical features.

A decision tree classifier with `max_depth=7` is trained using PyXAI's scikit-learn interface with a 80/20 train-test split. This depth produces trees with 78 leaves (Adult) and 57 leaves (German), achieving 85.98% and 82.50% test accuracy respectively. Averaged over all enumerated reasons, sufficient reasons contain 5.93 features (Adult) and 8.15 features (German), providing enough complexity for meaningful evaluation while remaining tractable for exact ground truth computation.

TABLE I
Dataset and Model Summary

	Adult Income	**German Credit**
Features	14	20
Instances	30,162	1,000
Tree depth	7	7
Leaves	78	57
Test accuracy	85.98%	82.50%
Mean ground truth reason size	5.93	8.15
Mean ground truth reasons/instance	20.6	39.6

B. Ground Truth Computation

All sufficient reasons per instance are enumerated using PyXAI's `sufficient_reason(n=Explainer.ALL)`, which implements the algorithm of Ignatiev et al. [2] using a Glucose SAT solver with blocking clauses. The computed reasons were verified against our own recursive tree-traversal sufficiency checker that fixes the subset features and explores both branches at every free-feature node. Agreement was 100% across 60 randomly selected test instances.

C. Evaluation Metrics

For each heuristic feature subset S, we identify the best-matching ground truth reason GT (the reason with highest overlap, ties broken by lowest waste) and compute the following metrics:

- **Sufficiency rate**: fraction of instances where S formally guarantees the prediction. Computed via tree traversal: fix features in S to instance values, let all others vary. If every reachable leaf predicts the same class, S is sufficient.
- **Overlap**: $|S \cap GT|/|GT|$, the fraction of ground truth features found by the heuristic.
- **Waste**: $|S \backslash GT|$, heuristic features absent from the ground truth reason.
- **Jaccard similarity**: $|S \cap GT|/|S \cup GT|$, a standard symmetric measure of set similarity that penalises both missed GT features and wasted heuristic features.
- **Repair cost**: starting from the heuristic subset S, the number of additional features (taken from the method's own ranking in descending importance order, skipping features already in S) that must be appended, one at a time, before S becomes sufficient. Sufficiency is re-checked by tree traversal after each addition, so the procedure uses only the model and the heuristic's ranking.

A sufficient reason specifies conditions such as "age $\leq$ 45.5," while LIME and SHAP produce feature importance scores without threshold information. We only compare whether a heuristic selected the feature "age," because LIME and SHAP cannot produce condition-level explanations. This means overlap, waste, Jaccard, and repair are computed at the feature level. The sufficiency rate metric is unaffected as it evaluates feature values against the tree's splits. This approach is consistent with prior evaluations by Ignatiev [3] and Salem [7].

Each instance has multiple valid sufficient reasons, and their order depends on the SAT solver's elimination sequence. We therefore compare each heuristic subset against the reason with highest overlap to avoid penalising heuristics that happen to select features from a different valid reason than the one returned first.

D. Experimental Setup

For each dataset, 100 correctly classified test instances are randomly sampled. For LIME and SHAP, the top-k features are selected for $k = 1, \ldots, 9$, where absolute value of the scores is used (as validated in Section III-E below). LIME is

configured with 1,000 perturbation samples and default kernel width (validated below). SHAP uses TreeExplainer for exact Shapley values as explained in Section II-B. Anchors uses the default precision threshold $\tau = 0.95$, though several precision thresholds are tested in Section III-E.

E. Alternative Translations and Sensitivity Experiments

Beyond top-k absolute value ranking, we test several alternative methods for converting heuristic scores into feature subsets. Each addresses a specific methodological concern or is grounded in prior work.

Signed vs. absolute ranking. Our main experiments use absolute value, which treats features opposing the prediction as equally important to fix. The alternative is positive-first ranking, which selects features that most support the predicted class. We test both for LIME and SHAP at $k = 1$–9 to determine which better aligns with formal sufficiency.

Cumulative SHAP. Rather than fixing k, select features in descending order of $|\phi_i|$ until their cumulative absolute Shapley value exceeds $p\%$ of the total $\sum_i |\phi_i|$. This is grounded in the efficiency property [5]: because Shapley values sum to $f(x) - E[f(x)]$, the cumulative fraction has a concrete interpretation as the proportion of the prediction's deviation from baseline that is accounted for. We test $p \in \{50, 70, 90\}$. This produces variable-size subsets that adapt to each instance's SHAP distribution. LIME coefficients do not satisfy the efficiency property and their sum has no interpretable meaning, therefore this translation method is only possible for SHAP.

Threshold SHAP. This method includes all features whose $|\phi_i|$ exceeds a fraction f of the maximum value $\max_j |\phi_j|$, inspired by the feature selection approach of Shaposhnikov et al. [12]. We test $f \in \{0.10, 0.25, 0.50\}$. Like cumulative SHAP, this produces variable-size subsets, but the selection is relative to the most important feature rather than the total contribution.

Formal Feature Attribution (FFA). Yu et al. [11] define a feature's formal importance as the fraction of all sufficient reasons in which it appears. We compute this from our enumerated ground truth reasons and use the resulting ranking to select the top k features. FFA uses ground truth information, which does not make it a fair comparison with LIME or SHAP but rather a formal ranking baseline for when we have exact knowledge of all sufficient reasons. Huang et al. [14] formalise the related concepts of feature necessity and relevancy, where a feature is relevant if it appears in at least one sufficient reason.

Anchors precision sweep. Anchors uses a precision threshold τ that controls how many sampled perturbations must agree with the prediction. The default is $\tau = 0.95$. To determine whether the low sufficiency rates of Anchors are caused by this threshold or by the sampling approach itself, we test $\tau \in \{0.80, 0.85, 0.90, 0.95, 0.99\}$.

LIME hyperparameter sensitivity. LIME's local linear model depends on two key parameters: `num_samples` (the number of perturbations generated, default 1,000) and `kernel_width` (the radius of the neighbourhood used for weighting, default $\sqrt{n} \times 0.75$ where n is the number of features). Salem [7] found `kernel_width` had negligible effect on synthetic data. We test whether this holds on real data by evaluating `num_samples` $\in \{100, 500, 1000, 5000\}$ and `kernel_width` $\in \{0.25, 0.75, \text{default}\}$ at $k = 5$.

IV. Results

A. Sufficiency Curve

Table II shows the sufficiency rate of LIME and SHAP as the feature budget k increases from 3 to 9, alongside FFA as the formal ranking baseline defined in Section III-E.

TABLE II
Sufficiency Rate (%) by Method and k

k	Adult Income			German Credit		
	LIME	SHAP	FFA	LIME	SHAP	FFA
3	0	0	2	1	1	0
4	13	3	52	2	3	0
5	35	31	72	3	12	14
6	66	**96**	87	3	28	35
7	74	**99**	93	6	36	62
8	85	**99**	97	9	44	79
9	89	**99**	100	19	58	90

Sufficiency rates are effectively zero at $k \leq 3$ on both datasets, in line with mean ground truth reason sizes of roughly six features on Adult Income and eight on German Credit. On Adult Income, LIME leads SHAP at $k = 5$ (35% vs 31%), but SHAP overtakes sharply at $k = 6$ (96% vs 66%) and plateaus at 99% from $k = 7$ onward, while LIME climbs more gradually to 89% at $k = 9$. SHAP surpasses the FFA baseline from $k = 6$ to $k = 8$ (for example 99% vs 93% at $k = 7$), with FFA edging ahead only at $k = 9$ (100% vs 99%). On German Credit both methods perform far worse: at $k = 7$ SHAP reaches 36% and LIME 6%, and even at the largest budget tested ($k = 9$) they reach only 58% and 19%. SHAP stays below FFA from $k = 5$ onward on this dataset (58% vs 90% at $k = 9$).

At the default precision threshold $\tau = 0.95$, Anchors produces short rules: on Adult Income, 81 of 100 instances receive a rule with only 1 or 2 features. Across all instances, Anchors achieves 4% sufficiency on both datasets, with the sufficient instances all having rule lengths of 17 or more features. The precision sweep in Section IV-C provides further detail.

B. Diagnostic Metrics

Table III reports overlap, waste, Jaccard similarity, and repair cost at two budgets. The first matches each dataset's mean reason size ($k = 6$ for Adult Income, $k = 8$ for German Credit), the smallest budget at which a sufficient subset is even possible and therefore a fair operating point per dataset. The second is $k = 9$, the largest budget we evaluate, which lets us compare both datasets at an identical generous budget.

On Adult Income, SHAP is already near-perfect at the mean reason size: at $k = 6$ it reaches 0.992 overlap with a repair cost of 0.07 features, and by $k = 9$ overlap is 0.999 and

TABLE III
DIAGNOSTIC METRICS AT A BUDGET MATCHING THE MEAN REASON SIZE AND THE LARGEST BUDGET EVALUATED ($k = 9$)

k	Method	Overlap	Waste	Jaccard	Repair
Adult Income (mean reason size 5.93)					
6	SHAP	0.992	0.35	0.936	0.07
	LIME	0.937	0.76	0.834	0.80
	FFA	0.981	0.15	0.957	0.23
9	SHAP	0.999	2.19	0.756	0.01
	LIME	0.983	3.00	0.660	0.22
	FFA	1.000	1.76	0.804	0.00
German Credit (mean reason size 8.15)					
8	SHAP	0.876	2.50	0.629	1.81
	LIME	0.752	3.34	0.495	6.45
	FFA	0.975	0.88	0.868	0.35
9	SHAP	0.915	3.02	0.628	1.25
	LIME	0.782	3.97	0.492	5.33
	FFA	0.989	1.27	0.850	0.14

repair 0.01. Jaccard peaks and then falls as the budget grows past the mean (0.936 at $k = 6$ versus 0.756 at $k = 9$), since features added beyond the reason are wasted and waste rises from 0.35 to 2.19. LIME trails SHAP throughout, with more waste and a higher repair cost: 0.80 against 0.07 features at $k = 6$ on Adult, and 6.45 against 1.81 at $k = 8$ on German. German Credit is worse across the board: at its mean reason size ($k = 8$) SHAP reaches only 0.876 overlap and still needs 1.81 repair features, and at $k = 9$ it is 0.915 overlap with 1.25.

C. Anchors Precision Sweep

Even at $\tau = 0.99$, Anchors achieves only 9% sufficiency on both datasets, with mean rule lengths of 4.7 (Adult) and 7.0 (German). Higher τ produces longer rules but does not substantially improve sufficiency.

D. Alternative Translations and Sensitivity

Signed vs. absolute ranking. At $k = 7$ on Adult Income, positive-first LIME achieves 3% sufficiency vs. 74% with absolute, and positive-first SHAP achieves 62% vs. 99%. On German Credit, positive-first LIME drops to 2% (from 6%) and positive-first SHAP to 14% (from 36%).

Cumulative SHAP. At 90% on German Credit, the mean subset size is 7.5 and sufficiency reaches 41%, compared to 36% for top-k at $k = 7$. On Adult Income, 90% cumulative gives a mean subset of 4.0 features (below the ground truth mean of 5.93), resulting in 0% sufficiency. At 50% and 70%, subsets are too small for sufficiency on either dataset.

Threshold SHAP. At $f = 0.10$ (include features with $|\phi_i|$ exceeding 10% of the maximum), the mean subset size is 4.3 on Adult and 5.8 on German, producing 10% and 30% sufficiency respectively. Sufficiency increases with subset size: on German Credit, instances receiving 8 or more features reach 66.7% sufficiency at $f = 0.10$. At $f = 0.25$, mean subset sizes drop to 2.5 (Adult) and 3.2 (German), with sufficiency at 0% and 7%. At $f = 0.50$, subsets average 1.5 and 1.9 features with 0% sufficiency on both datasets.

LIME sensitivity. Varying `kernel_width` from the default (≈ 2.8) to 0.25 drops sufficiency from 35% to 2% at $k = 5$ on Adult Income. Varying `num_samples` from 100 to 5,000 shows 9% at 100, 33% at 500, 35% at 1,000, and 42% at 5,000. On German Credit, both non-default kernel widths (0.25 and 0.75) result in 0% sufficiency at $k = 5$, compared to 3% with the default.

V. DISCUSSION

A. SHAP vs. LIME

The results showed SHAP outperforming LIME at budgets above the mean ground truth reason size on both datasets. There are structural differences between LIME and SHAP that explain these findings. SHAP computes marginal contributions across all feature coalitions, which accounts for feature interactions by assigning value to both features when they jointly determine the prediction. LIME fits a single local linear model that assumes features contribute independently, and can rank one interacting feature high while missing the other. SHAP's ranking is also deterministic and nested: the features at $k = 7$ for a given instance are always $k = 6$ plus one more. LIME refits the Lasso for each k with fresh perturbations, so rankings can shift between k values. These properties explain the crossover observed at $k = 6$ on Adult Income: SHAP's ranking placed the remaining critical features at positions 5–6, while LIME's refitting did not consistently place them there. The LIME sensitivity experiment reinforces this: varying `kernel_width` from the default to 0.25 dropped sufficiency from 35% to 2%, confirming that LIME's local approximation is sensitive to neighbourhood configuration in ways that SHAP's exact tree traversal is not.

These findings are broadly consistent with prior work, although the comparison is not exact. Ignatiev [3] found SHAP more often correct than LIME on three of five datasets (Table 1 in [3]); on Adult Income, however, his evaluation reports the opposite ordering (LIME 30.8% vs. SHAP 19.1% correct), whereas we find SHAP ahead of LIME at high k. Ignatiev measures subset-minimal correctness of a single explanation on boosted trees, while we measure sufficiency of a top-k subset on single decision trees, so the two are not directly comparable. Salem [7] likewise found SHAP aligns well with formal properties on synthetic data. Across both, the pattern is the same: SHAP's deterministic ranking aligns with formal sufficiency more reliably than LIME's, while at low k (below the ground truth reason size) neither achieves meaningful sufficiency. The kernel-width finding contradicts Salem's observation that `kernel_width` had negligible effect on synthetic data, consistent with Slack et al. [10] that LIME's perturbation distribution does not match the data distribution on real datasets.

B. Anchors

Anchors sufficiency rates remained near 0%. There are several structural reasons for Anchors low sufficiency rates:

Anchors probabilistic precision checks the sampled perturbations rather than all possible feature assignments, while sufficient reasons require the prediction to hold for *all* possible values of free features. Regardless of τ, the gap allows corner cases that the sufficiency checker catches. The second reason is that Anchors beam search stops early. On Adult Income, 81% of instances receive rules with length 1 and 2 features because short rules easily achieve the precision threshold on the majority class. The precision sweep experiment in Section IV-C confirms the gap was not a matter of threshold tuning. Ignatiev [3] similarly observed discrepancies between Anchors' reported precision and precision measured via formal methods.

C. Dataset Dependency

Every method performs worse on German Credit (20 features) than on Adult Income (14 features). With 20 features there are $\binom{20}{7} = 77{,}520$ possible 7-feature subsets compared to $\binom{14}{7} = 3{,}432$ for Adult. The heuristics must identify roughly eight features, the mean German reason size, which the budgets we test only barely reach, from a much larger pool. The FFA baseline illustrates this: even with exact knowledge of all sufficient reasons, frequency-based ranking reaches only 62% on German Credit at $k = 7$ (90% at $k = 9$) compared to 93% (100%) on Adult. On Adult, SHAP matches or exceeds FFA from $k = 6$ to $k = 8$, which suggests that instance-specific Shapley values can find the right feature combination even when global frequency ranking does not; FFA overtakes only at $k = 9$. On German Credit, SHAP stays well below FFA from $k = 5$ onward (58% vs 90% at $k = 9$), indicating that the higher dimensionality limits SHAP's ability to rank all critical features in the top positions. German Credit's mean reason size of roughly eight features sits at the very top of the budgets we test, and even at that budget sufficiency remains far below what the same methods reach on Adult Income.

D. Repair Feasibility

The repair cost in Table III addresses RQ3 directly: starting from the top-k subset, features are added in the method's own ranking order and sufficiency is re-checked after each addition. On Adult Income, SHAP's repair cost is negligible once the budget reaches the mean reason size (0.07 features at $k = 6$, 0.01 at $k = 9$), because the top-k subset is already sufficient for almost all instances and any shortfall is closed by one or two further SHAP-ranked features. On German Credit the cost is far higher (1.81 features for SHAP at $k = 8$, 1.25 at $k = 9$, and 6.45 and 5.33 for LIME), since the next-ranked features are often not the ones needed to make the subset sufficient. Repair cost therefore follows the same pattern as sufficiency: cheap for SHAP on the moderate-dimensional dataset, rising sharply with dimensionality, and worse for LIME than SHAP at every budget.

E. Limitations

This work has several limitations that should be considered when interpreting the results. Integer encoding treats categorical features as ordinal, meaning the tree's threshold splits group alphabetically adjacent categories together without semantic meaning (discussed in Section III-A). Only a single tree depth (`max_depth=7`) was tested. Shallower trees would produce fewer and shorter sufficient reasons while deeper trees would produce more and longer ones, and the relationship between tree depth and sufficiency rates remains unexplored. We evaluate 100 instances per dataset. The observed effect sizes are large enough (0% vs 99%) that statistical significance is not a primary concern, but the sample may not capture rare edge cases. As discussed in Section III-C, diagnostic metrics such as overlap and waste operate at the feature level rather than the condition level, though the sufficiency rate itself is exact. LIME's results are subject to randomness from its perturbation sampling, and the reported sufficiency rates can shift by several percentage points across runs and random seeds. Repair cost is measured along each method's own importance ranking. For LIME, whose per-k Lasso fits are not nested, the repair uses a single all-feature LIME fit to supply a fixed ranking; for Anchors, which does not emit a ranked feature list, repair is undefined and therefore not reported. The reported costs are specific to these ranking choices. Finally, all results are specific to single decision trees and may not generalise to neural networks or ensemble models, where ground truth computation requires more expensive SAT or MILP encodings [2].

VI. Conclusions

This paper evaluated the extent to which LIME, SHAP, and Anchors can satisfy formal sufficiency criteria from logic-based explanation frameworks. The main finding is that heuristic methods can produce formally sufficient feature subsets, but only when the feature budget exceeds the ground truth reason size and on datasets with moderate feature dimensionality.

RQ1 asked how the heuristic outputs can be translated into feature subsets. Of the translations tested, top-k selection by absolute score is the most reliable. It beats signed ranking decisively (74% versus 3% for LIME at $k = 7$ on Adult Income), since features that oppose the prediction still need to be fixed for the guarantee to hold. The variable-size alternatives, cumulative and threshold SHAP, only match top-k when they happen to select a comparable number of features, and do not improve on it.

RQ2 asked to what extent the resulting subsets satisfy sufficiency. SHAP's ranking does so more reliably than LIME's, particularly at $k \geq 6$ where SHAP's deterministic and nested ranking captures the remaining critical features: SHAP reaches 99% sufficiency by $k = 7$ on Adult Income, while LIME climbs to 89% only by $k = 9$. Sufficiency is strongly dataset-dependent, however. On German Credit even a nine-feature budget yields just 58% for SHAP and 19% for LIME, and Anchors stays near zero on both datasets regardless of its precision threshold.

RQ3 asked what it costs to repair an insufficient explanation. The cost is low for SHAP on Adult Income but grows with dimensionality. At a budget matching the mean reason size,

SHAP needs a mean of 0.07 additional features on Adult Income ($k = 6$) and 1.81 on German Credit ($k = 8$), while LIME needs 0.80 and 6.45. At the largest budget tested ($k = 9$) the Adult cost falls to near zero (0.01 for SHAP, 0.22 for LIME) while the German cost stays substantial (1.25 for SHAP, 5.33 for LIME). Because the sufficiency check requires only the model, this repair step is practical to deploy for SHAP on the moderate-dimensional dataset, but its cost rises sharply on the higher-dimensional one.

Future work should investigate one-hot encoding with proper mutual-exclusivity constraint handling following Gorji and Rubin [9], varying tree depth to assess how it affects sufficiency rates and reason counts, and experiments with models such as random forests and gradient-boosted trees where ground truth computation requires more expensive SAT or MILP encodings [2]. Sufficiency checking that restricts free features to realistic values rather than all possible assignments would also bring the pipeline closer to practical deployment conditions.

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